



Sequel Detective

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FACULTY CAPSTONE PROJECT

7/16/2026



NEUMONT
UNIVERSITY



Problem

- Worksheet-style SQL
- Little investigative context
- Syntax feels like the finish line

Goal

- Case-driven inquiry
- Evidence-backed reasoning
- SQL becomes an investigation tool

The question shifts from

"How do I write a SQL query?"

to

"What does the data prove?"

How The Project Evolved

The core idea stayed the same: students learn SQL by investigating evidence.

1 Analog Classroom Assignment



2 AI-Integrated Capstone Concept



3 Deterministic Web Implementation



This version keeps the investigation, but moves authority into deterministic services.



Learning Experience Is Evidence-Led

Students solve the case by:

- Asking questions
- Testing ideas
- Explaining what the data shows
- **Start with a case question**
- **Explore the database for clues**
- **Use SQL to test each lead**
- **Read results like evidence**
- **Build a conclusion they can defend**

The Student Steps Into The Case

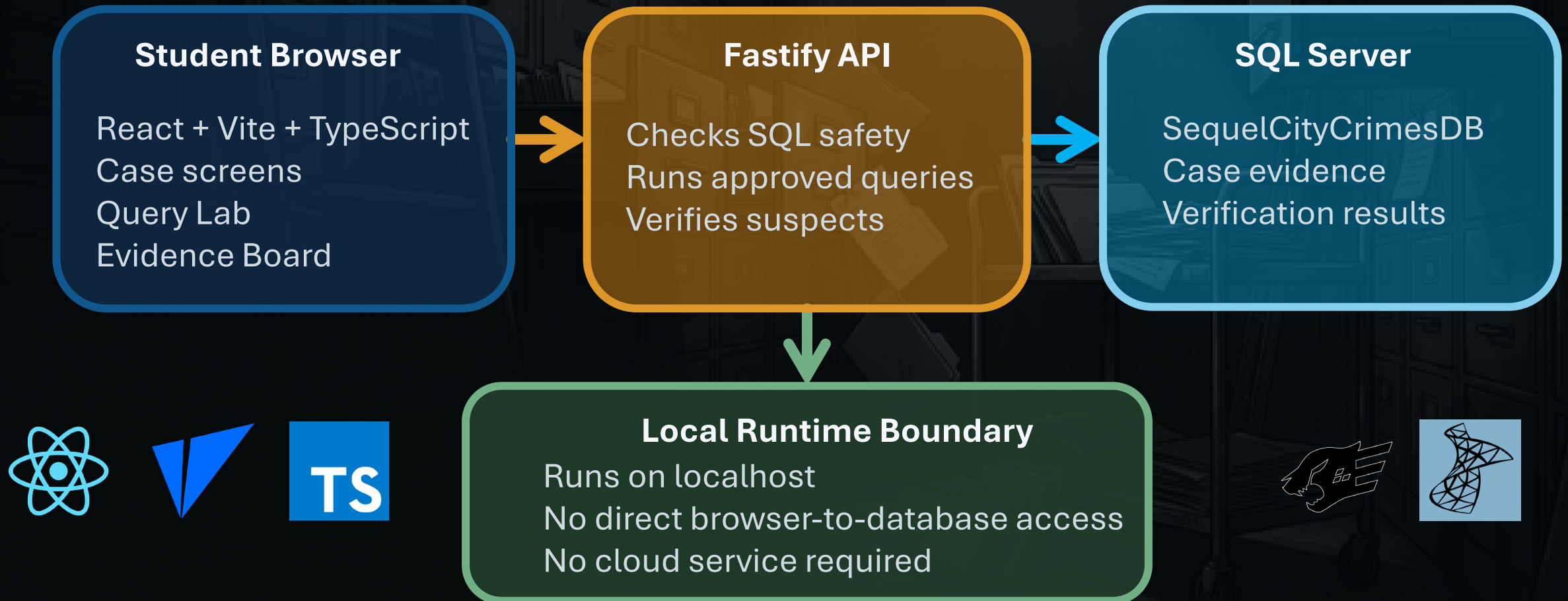
Each screen gives the student a reason to ask the next question.

- Choose the case
- Follow the briefing
- Query for leads
- Build the evidence board
- Test the theory



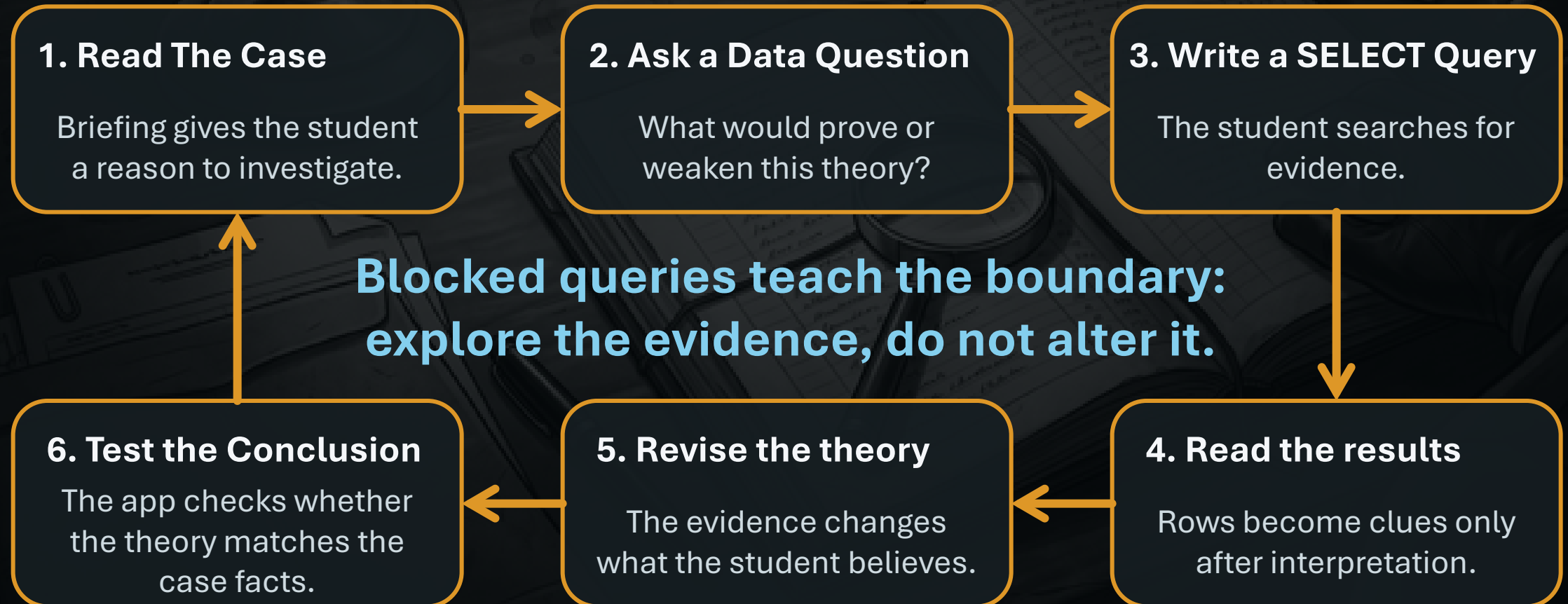
The App Keeps Evidence Local And Verifiable

The browser guides the investigation, but the backend and database decide what is safe and what is true.



Students Follow the Evidence

Each query is part of a loop: ask, test, read, revise.



Demonstration



AI Assisted Engineering Process

Codex implemented scoped work packages;
Gemini and AntiGravity audited against the SSOT.

Codex

- SSOT Creation
- Work Package Creation
- Scoped Implementation
- Mechanical Edits



Gemini / AntiGravity

- Independent audit
- SSOT alignment checks
- Scope and regression review
- Deterministic-boundary validation
- Cross-checking and verification



Human Developer

- Final decisions
- Acceptance review
- Instructional judgment
- Project direction



Successes And Challenges

The project worked best when the learning experience stayed grounded in evidence.

Successes

- Local-first browser runtime
- Deterministic SQL safety boundary
- Playable Case investigation
- Database-backed suspect checks

Challenges

- Turning a static assignment into a guided investigation
- Balancing support with student agency
- Blocking unsafe SQL without blocking exploration
- Hardening local SQL Server setup

**Disciplined scope kept the project coherent:
deterministic services first, narrative polish second.**

Future Plans

- Expand the case library
- Persist notebook and case state
- Build Instructor analytics
- Harden the Student Package
- Classroom deployment

Sequel Detective teaches SQL as evidence-based reasoning, not answer hunting.



Sequel Detective Statistics

Tracked files:	671
Markdown docs:	277
TypeScript files:	103
Documentation lines:	62,153
TypeScript lines:	28,851
SQL script lines:	40,976
Automated test files:	45

Build status: **PASSING** (web + API)

Test status: **PASSING** (web 179 + API suite)

All artwork generated by Gemini/Nano Banana



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QUESTIONS?

